Page No.
Date

Q Solve $12x \equiv 21 \pmod{39}$, $\gcd(12, 39) = 3$

So, $4x \equiv 7 \pmod{13}$, now $\gcd(4, 13) = 1$

$\therefore 4x_1 + 13y_1 = 1$

Chinese Remainder Theorem

⇒ In many situations, it is useful to break a congruence mod n into a system of congruences mod factors of n.

Ex Let a number x satisfies $x \equiv 25 \pmod{42}$

So $x = 25 + 42k$ for some integer k

We know, $42 = 7 \times 6$

So,

$\Rightarrow x = 25 + 7(6k)$

$= 25 + 6(7k)$

$x \equiv 25 \equiv 4 \pmod{7}$

Therefore $x \equiv 25 \pmod{42} \Rightarrow \begin{cases} x \equiv 4 \bmod 7 \\ x \equiv 1 \bmod 6 \end{cases}$

Theorem

⇒ Suppose $\gcd(m, n) = 1$

Given integers a and b, there exists exactly one solution $x \pmod{mn}$ to the simultaneous congruences.

$\Rightarrow x \equiv a \pmod{m}$, $x \equiv b \pmod{n}$

Page No.
Date

Date : 16 September 2026

1. Use Euclidean Algorithm to find the GCD of 252 and 105.

Soln

$252 = 105 \times 2 + 42$
$105 = 42 \times 2 + 21$
$42 = 21 \times 2 + 0$

So, $\gcd(252, 105) = 21$

2. Use extended euclidean algorithm
$56x + 15y = \gcd(56, 15)$

i	q_{i-1}	r_i	s_i	t_i
0	—	56	1	0
1	—	15	0	1
2	$56 \div 15 = 3$	$56 - 3 \times 15 = 11$	$1 - 3 \times 0 = 1$	$0 - 3 \times 1 = -3$
3	$15 \div 11 = 1$	$15 - 1 \times 11 = 4$	$0 - 1 \times 1 = -1$	$1 - 1 \times (-3) = 4$
4	$11 \div 4 = 2$	$11 - 2 \times 4 = 3$	$1 - 2 \times (-1) = 3$	$-3 - 2 \times (4) = -11$
5	$4 \div 3 = 1$	$4 - 1 \times 3 = 1$	$-1 - 1 \times 3 = -4$	$4 - 1 \times (-11) = 15$
6	$3 \div 1 = 3$	$3 - 1 \times 3 = 0$	$3 - 3 \times (-4) = 15$	$-11 - 3 \times (15) = -56$

$r_k = a \cdot s_k + b t_k$

$\Rightarrow (-4) \times 56 + (15) \times 15 = 1$

s_k $\quad$ t_k

$s_k = -4$
$t_k = 15$

Page No.
Date

3. Using Extended Euclidean Algorithm, find the modular inverse of 7 modulo 26.

i	q_{i-1}	r_i	s_i	t_i
0	–	26	1	0
1	–	7	0	1
2	$26 \div 7 = 3$	$26 - 3\times7 = 5$	$1 - 3\times0 = 1$	$0 - 3\times1 = -3$
3	$7 \div 5 = 1$	$7 - 1\times5 = 2$	$0 - 1\times1 = -1$	$1 - 1\times(-3) = 4$
4	$5 \div 2 = 2$	$5 - 2\times2 = 1$	$1 - 2\times(-1) = 3$	$-3 - 2\times4 = -11$
5	$2 \div 1 = 2$	$2 - 2\times1 = 0$	$-1\times2\times3 = 5$	$4 - 2\times(-11) = 26$

$r_k = a s_k + b t_k$

$\Rightarrow 1 = 26 \times (3) + 7 \times (-11)$

(s_k = 3, t_k = −11)

Inverse cannot be negative. Hence $(-11) = 15$

So,

$s_k = 3$

$t_k = 15$

So, modular inverse of 7 modulo 26 = 15

4. Modular inverse of 17 modulo 43

i	q_{i-1}	r_i	s_i	t_i
0	–	43	1	0
1	–	17	0	1
2	$43 \div 17 = 2$	$43 - 2\times17 = 9$	$1 - 2\times0 = 1$	$0 - 2\times1 = -2$
3	$17 \div 9 = 1$	$17 - 1\times9 = 8$	$0 - 1\times1 = -1$	$1 - 1\times(-2) = 3$
4	$9 \div 8 = 1$	$9 - 1\times8 = 1$	$1 - 1\times(-1) = 2$	$-2 - 1\times3 = -5$
5	$8 \div 1 = 8$	$8 - 1\times8 = 0$		

So, $\Rightarrow 43\times(2) + 17\times(-5) = 1$

So, modular inverse is $\Rightarrow 43 - 5 = 38$

Page No.
Date

RSA

1. Generating Key - Two large prime No.'s p & q

$$n = p \times q$$
$$\phi(n) = (p-1) \times (q-1)$$

Choose e, $1 \leq e \leq \phi(n)$, $gcd(e, \phi(n)) = 1$
$d * e = 1 \bmod (\phi(n))$

Public Key = (n, e), Private Key = (n, d)

2. Encryption $\Rightarrow$ Ciphertext (c) = $m^e [\bmod(n)]$
3. Decryption $\Rightarrow$ Message (m) = $c^d [\bmod(n)]$

Q. p = 3, q = 11. Calculate:

a) n b) $\phi(n)$ c) Choose a valid public exponent e

a) $n = p \times q$
$= 3 \times 11 = 33$

b) $\phi(n) = (3-1) * (11-1)$
$= 2 \times 10$
$= 20$

c) Choose e, $1 \leq e \leq \phi(n)$
$1 \leq e \leq 20$

If e = 3, then ~~gcd~~
$gcd(e, \phi(n)) = 1$

So, e = 3

Page No. _____
Date

d) $d * e = 1 \mod \phi(n)$

$3d = 1 \mod 20$

so, $\underline{d = 7}$

e) Public Key = (33, 3)

Private Key = (33, 7)

Q Using the RSA keys obtained from Q5, encrypt the message $m = 4$, and then decrypt the ciphertext to recover the original message. Show all modular arithmetic steps.

Soln

$c = m^e \ [\mod(n)]$

$= 4^3 \ [\mod 33]$

$= 64 \mod 33$

$\underline{c = 31}$

$m = c^d \ [\mod(n)]$

$= 31^7 \mod 33$

$\underline{m = 4}$